A Method for Active Marine Target Detection Based on Complex Interferometric Dissimilarity in Dual-Channel ATI-SAR Systems

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***Abstract*— Synthetic aperture radar (SAR) operating in an along-track interferometric (ATI) mode has the advantage of minimum detectable velocity (MDV) in active marine target detection. However, most of the conventional ATI detectors fail to identify the marine targets cruising with blind speeds, in which case the interferometric phases of these targets are closely distributed around those of the sea background due to phase wrapping and aliasing through ATI processing. To address this issue, we propose a method to detect the active marine targets based on complex interferometric dissimilarity for dual-channel ATI-SAR systems. First, an interferometric bilateral filter is designed to smooth the random noises and locally adapt to the spatial structure of the interferogram for measuring the interferometric magnitude and phase. Then, the target detection metric is constructed based on the complex interferometric dissimilarity between the marine targets and the sea background. By adaptively regressing into a magnitude-based test toward the blind-speed targets, the target detection metric can mitigate the blind-speed detection problem and thus yield a satisfactory detection result. Furthermore, this metric is of a constant false-alarm rate (CFAR), and its probability density function (pdf) is derived to facilitate the detection threshold computation. Finally, both the simulated and real-data processing results are given to validate the superiorities of the proposed method.**

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***Index Terms*— Active marine target detection, along-track interferogram (ATI), blind speeds, interferometric bilateral filter, synthetic aperture radar (SAR).**

I. Introduction

SYNTHETIC aperture radar (SAR) has been playing an important role in maritime affairs, such as traffic control, fishery management, and vessel search and rescue [1], [2], due to its all-weather and all-day imaging capability. Since active marine target detection is necessary in these applications, it has been drawing much attention.

Single-channel SAR mainly takes the advantage of intensity information of SAR images to detect marine targets. Under a low angle of incidence, the normalized radar cross section (NRCS) of ships is usually higher than that of the sea surface in SAR images because of different microwave backscattering characteristics between ships of metal construction and seawater [3]–[6]. However, as the SAR image resolution increases, new challenges, such as distributed characteristics of targets, enhanced contrast of coherent speckles, and distinct texture of sea waves, and traditional problems, such as bright wakes of active ship and leaping fish [3], and faint intensity differences between targets and clutter, will increase the probability of missed detection of targets and cause a high false-alarm rate. With the development of the techniques, target detection performance in single-channel SAR systems has been greatly improved. The local intensity variation [7], [8], [11] and ship spatial structure features [9], [10] are exploited to enhance the target-to-clutter contrast. However, isolated bright clutter may be falsely detected as targets during the processing. In order to remove the isolated false alarms, a method based on shape features of the ships is proposed [12], and, in [13], particular ship features, such as length, width, and aspect ratio, were used to form a ship discrimination framework [13]. In addition, a hierarchical saliency filtering method is proposed in [14], which is based on the multilayer selective cognition property of the human visual system. These methods usually allow an enhanced detection result for the large-size marine targets in the complex background by exploiting the intelligence information of human sense, but they may not be suitable for the targets with small size and low signal-to-clutter ratio (SCR).

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Moving target detection (MTD) for the multi-channel SAR [15]–[18], such as imaging space-time adaptive processing (ISTAP), displaced phase center antenna (DPCA), Kelly's generalized likelihood ratio test (GLRT), adaptive matched filter (AMF), and a class of two-stage detections, can adaptively suppress clutter while reserving the interested targets. Thus, these methods can greatly improve the target-to-clutter contrast and yield more satisfactory results compared with the detection in single-channel SAR systems.

As a special case of multi-channel SAR, along-track interferometric SAR (ATI-SAR) employs two channels separated by a short baseline along the platform track to record the echoes reflected by a scene [19]. The phase change of the two SAR images for a scene is proportional to the Doppler shift of the backscattered signal and, thus, to the line-of-sight target velocities [26], [27]. ATI-SAR systems have been initially used in measuring slowly moving phenomena such as ocean currents and debris flows. Since ATI-SAR systems have the advantage of low minimum detectable velocity (MDV) and being economical and space saving, they have been widely applied in MTD for the current and near-future surveillance satellites such as RADARSAT-2 (RS2) [20], [22], TerraSAR-X (TSX) or TanDEM-X (TDX) [19], [23], and AIRSAR [24], [25]. Thanks to the extensive MTD experiments on the real ATI-SAR systems, the techniques have been developed rapidly. The compensation of system's response errors [28], [29], registration and calibration of dual-channel images [20], [34], modeling realistic clutter and target characteristics [4], [30]–[32], imaging high-resolution SAR imageries [33], [4], and detecting targets using multi-domain features [20]–[22] can much improve the MTD performance for ATI-SAR systems.

For marine MTD based on ATI techniques, many valuable investigations have been made. Considering that random and inevitable thermal noises cause estimation bias in ATI processing, the complex multi-look interferogram is proposed to smooth the noises by averaging independent and identically distributed samples in the complex interferogram [20]. With this multi-look method, ATI detection and the fusion of DPCA and ATI can significantly improve the MTD performance [22]. The metric by joining the interferometric magnitude and phase (IMP) has been proposed to improve SCR and for the convenience of detecting ground moving targets in dual-channel ATI-SAR systems [35], whereas the IMP method cannot be directly used for active marine target detection, because sea clutter has temporal and space-dependent velocities, which differs from that of the stationary ground. Then, a method by combining sea's interferometric magnitude and phase (SIMP) is developed exactly for the active marine target detection in dual-channel ATI-SAR systems [36], [37]. In addition, the geometrical perturbation–polarimetric notch filter (GP–NF) in [58], used in quad- or dual-polarization SAR systems, is recently extended to dual-channel ATI-SAR systems for marine target detection. In this case, the GP–NF is equivalent to a simplified detection form, called as the interferometric notch filter (INF) [38]. Because the foregoing SIMP, GP–NF, and INF approaches are able to suppress signals reflected from the sea background by designing the notch-adaptive filters in the interferometric phase domain, they can obtain enhancements in SCR for the targets with different interferometric phases from that of the sea clutter [36]–[38]. However, due to the phase wrapping and aliasing through ATI processing, the interferometric phases for moving targets cruising with some specific speeds would be closely adjacent or even equal to the interferometric phase of the sea clutter. At this time, ATI detection, the SIMP method, and the INF fail to work adequately. A graphical explanation of the filter responses of the SIMP method, i.e.,$(1-\cos(\varphi-\varphi_s))$, against the interferometric phases, i.e., φ, is shown in Fig. 1, where it assumes that the interferometric phase is suppressed at 0 rad in the interferometric phase filter, i.e., $\varphi_s = 0$ rad. The moving targets in the blind-speed zones shown in Fig. 1, corresponding to those with the interferometric phases near $\varphi_s + 2k\pi$, $k \in \{0, \pm1, \pm2, \ldots\}$, are greatly suppressed in the filtering; therefore, they become undetectable in the sea background using the SIMP method. This blind-speed problem also presents in the ATI and INF methods. The radar system with multiple carrier frequencies [39], [40] and nonuniform linear subarrays [41]–[43] can solve the blind-speed problem, but these solutions are more complex in system implementation and signal processing, which are beyond the scope of our study in this article.

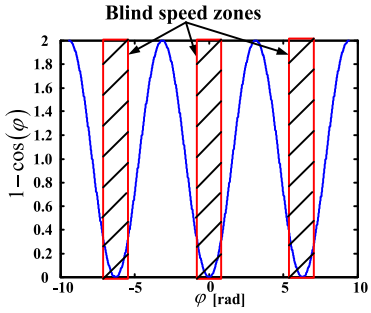


Fig. 1. Filter response of SIMP in the interferometric phase domain.

Following the previous work, we propose a new method to detect active marine target in a dual-channel and single-frequency ATI-SAR system to mitigate the blind-speed problem. An interferometric bilateral filter is first designed to smooth the noisy speckles and locally adapt to the image spatial structure during ATI measurements. Since then, marine targets and sea clutter are easy to be distinguished in the complex interferogram by comparing the magnitude and the phase. The target detection metric is constructed by measuring the complex interferometric dissimilarity with respect to the sea background based on the exponential-kernel function. By utilizing the information of both the IMP, the target detection metric can dramatically improve the target-to-clutter contrast. Moreover, this metric can adaptively regress into a magnitude-based test for the targets with blind speeds and thus ease the blind-speed problem. The probability density function (pdf) of the target detection metric is derived, which will facilitate the detection threshold computation with a constant false-alarm rate (CFAR) property in the varied sea backgrounds. Finally, both simulated and real-data processing results are provided to validate the superiorities of the proposed method.

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The remainder of this article is organized as follows. Section II establishes the signal model of the complex interferogram. Section III introduces the principle of the proposed method and its CFAR detection scheme in detail. Section IV exhibits both real-data and simulated processing results, and gives extensive discussion to validate the proposed method. Finally, conclusions are drawn in Section VI.

II. Signal Model of the Complex Interferogram

In dual-channel ATI-SAR systems, radar echoes reflected from a scene are received by two antennas equipped along the platform track. After SAR azimuth-range imaging, motion compensation, and image registration and calibration [24], dual-channel SAR images can be acquired with a good match. Let $\mathbf{Z}(k) = [z_1(k), z_2(k)]^{\mathrm{T}}$ collect the complex data of the images at the same coordinate, where $z_i(k)$ denotes the kth single-look SAR image of the channel i, $i \in \{1, 2\}$, and the superscript T represents the transpose operator.

The correlation relationship of these two images is measured by the sample covariance matrix, i.e., $\mathbf{R}$, which is acquired by averaging n independent and identical distributed samples [20], [21], and formulated as follows:

$$\mathbf{R} = 1/n \sum_{k=1}^{n} \mathbf{Z}(k)\mathbf{Z}(k)^H = \begin{bmatrix} 1/n \sum_{k=1}^{n} |z_1(k)|^2 & 1/n \sum_{k=1}^{n} z_1(k)z_2(k)^* \\ 1/n \sum_{k=1}^{n} z_1(k)^* z_2(k) & 1/n \sum_{k=1}^{n} |z_2(k)|^2 \end{bmatrix} \tag{1}$$

where the superscripts H and $*$ represent the conjugate complex transpose and the complex conjugate, respectively.

In (1), the off-diagonal element $1/n \sum_{k=1}^{n} z_1(k)z_2(k)^*$ of $\mathbf{R}$ indicates the complex n-look interferogram. The normalized complex n-look interferogram is therefore defined as follows [20]:

$$I_n = \frac{1/n \sum_{k=1}^{n} z_1(k)z_2(k)^*}{\sqrt{E[|z_1|^2]E[|z_2|^2]}} = \xi e^{j\varphi} \tag{2}$$

where $|\cdot|$ represents the modulus of a complex scalar, $\varphi = \arg(I_n)$ denotes the complex n-look interferometric phase, and ξ denotes the normalized complex n-look interferometric magnitude. In [38], it has proved that the interferometric magnitude performs better in ship detection than single-channel amplitude/intensity based on the SCR improvement [38]. The interferometric phase is proportional to the radial velocity of the backscattered signal [19] and is expressed as follows:

$$\varphi = \left[\frac{2\pi}{\lambda} 2v_r \frac{d}{V_a}\right]_{-\pi,\pi} \tag{3}$$

where v_r denotes the radial velocity of the backscattered signal, λ, d, and V_a denote the radar wavelength, the effective baseline length [19], and the platform velocity along the track, respectively, and $[\cdot]_{-\pi,\pi}$ represents the periodic phase ambiguity in the range of $[-\pi, \pi]$ rad after ATI processing.

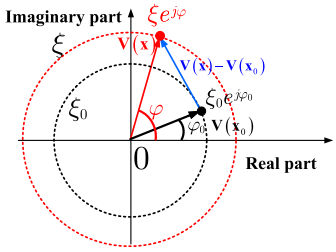


Fig. 2. Complex interferometric plane.

III. Proposed Detection Method

A. Principle of the Proposed Method

1) Interferometric Bilateral Filter: The radar echoes inevitably contain thermal noises, and this condition would affect the estimation accuracy of the complex interferometric information. The complex multi-look processing is a popular way to reduce the noises by averaging samples in the rectangular spatial window. However, this way leads to a sever reduction in image resolution and causes estimation errors for the objects with irregular interferogram. Bilateral filter uses the spatial distance and radiometric similarity to locally adapt to the image spatial structure to restore the image edges while smoothing noises in gray-level images [44], [45]. In fact, the spatial structure of imaged objects in the interferogram is significantly informative but has been not exploited in the interferometric measurements up to now.

Motivated by the relevant work, an interferometric bilateral filter is designed to estimate the complex interferogram based on the spatial structure and complex interferometric information of imaged objects. In particular, assume that 2-D location of a pixel is denoted by $\mathbf{x} = [x, y]^{\mathrm{T}}$, where x and y are the range and azimuthal coordinates, respectively. The complex interferometric plane is shown in Fig. 2, where the complex interferogram of the pixel, i.e., the interferometric magnitude ξ and the interferometric phase φ, can be denoted by the interferometric vector $\mathbf{V} = [\xi\cos(\varphi), \xi\sin(\varphi)]^{\mathrm{T}}$. With respect to a given reference pixel, e.g., the pixel with a 2-D location of $\mathbf{x}_0$, the interferometric bilateral filter is executed as follows:

$$\hat{\mathbf{V}}(\mathbf{x}_0) = \frac{1}{S} \int_{\mathbf{x}\in\Omega} \mathbf{V}(\mathbf{x}) f_s(\|\mathbf{x} - \mathbf{x}_0\|) f_r(\|\mathbf{V}(\mathbf{x}) - \mathbf{V}(\mathbf{x}_0)\|) d\mathbf{x} \tag{4}$$

with the normalized coefficient

$$S = \int_{\mathbf{x}\in\Omega} f_s(\|\mathbf{x} - \mathbf{x}_0\|) f_r(\|\mathbf{V}(\mathbf{x}) - \mathbf{V}(\mathbf{x}_0)\|) d\mathbf{x} \tag{5}$$

where $\|\mathbf{x} - \mathbf{x}_0\|$ denotes the spatial distance of the pixel $\mathbf{x}$ from the reference pixel $\mathbf{x}_0$, $\|\mathbf{V}(\mathbf{x}) - \mathbf{V}(\mathbf{x}_0)\|$ denotes the norm of their interferometric vector distance, as shown in Fig. 2, the operator $\|\cdot\|$ is the L_2 norm, f_s and f_i denote the weighing functions of the spatial distance and the complex interferometric similarity, respectively, and Ω denotes the neighborhood union of the reference pixel in 2-D location.

Generally, f_s and f_r have to be chosen to attain a maximum in zero and tend to zero if their arguments go to infinity.

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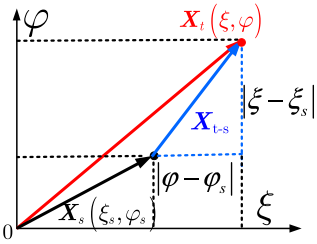


Fig. 3. 2-D feature plane of the IMP.

For example, the two functions are selected following the approach in [44], that is,

$$f_s(u)=\exp\left(-u^2/h_s^2\right),\quad f_r(u)=\exp\left(-u^2/h_r^2\right) \tag{6}$$

where the parameter h_s controls the spatial extent of the filtering and the parameter h_r controls the amount of the filtering in terms of the complex interferometric proximity.

In the perspective of the practical application, the interferometric bilateral filter is applied by

$$\hat{\mathbf{V}}(\mathbf{x}_0)=\sum_{\mathbf{x}_i\in\Omega} w_i\mathbf{V}(\mathbf{x}_i) \tag{7}$$

where the neighborhood union Ω can be set to $n\times n$-looks in the range and azimuthal directions, just as the window condition in the conventional multi-look process. After performing the preselection of pixels, the weighting w_i is

$$w_i=\frac{f_s(\|\mathbf{x}_i-\mathbf{x}_0\|)f_r(\|\mathbf{V}(\mathbf{x}_i)-\mathbf{V}(\mathbf{x}_0)\|)}{\sum_{\mathbf{x}_i\in\Omega} f_s(\|\mathbf{x}_i-\mathbf{x}_0\|)f_r(\|\mathbf{V}(\mathbf{x}_i)-\mathbf{V}(\mathbf{x}_0)\|)}. \tag{8}$$

With the interferometric vector $\hat{\mathbf{V}}(\mathbf{x}_0)$ estimated, the normalized interferometric magnitude is the vector's modulus, i.e., $\xi=|\hat{\mathbf{V}}(\mathbf{x}_0)|$, and the corresponding interferometric phase is the rotation angle from the "Real part" of axis to the interferometric vector shown in Fig. 2, i.e., $\varphi=\langle\hat{\mathbf{V}}(\mathbf{x}_0),[\,1\ ,\ 0\,]^{\mathrm{T}}\rangle$.

2) Principle of the Detection Metric: After applying the interferometric bilateral filter, the estimates of the IMP are acquired. The 2-D interferometric feature plane is shown in Fig. 3. A target pixel with the interferometric magnitude ξ and phase φ can be denoted by the interferometric feature vector $\mathbf{X}_t=[\,\xi,\ \varphi\,]^{\mathrm{T}}$. As to the sea background, the interferometric phase of every pixel corresponds in essence to the radial velocity of the sea surface at that point. In a specific ocean region, the sea surface is modeled as an object moving at a nearly uniform velocity [19]. The average value of the interferometric phase for the sea clutter, i.e., φ_s, corresponds to the average velocity of sea surface, and, meanwhile, a higher average velocity of sea surface means a larger φ_s in absolute value and a possibly higher sea current [36]. The interferometric magnitude depends on the integrated results of the point's backscattering properties and the dual-channel correlation. The average value of the interferometric magnitude for the sea clutter, i.e., ξ_s, approximately stands for the key magnitude characteristics of the sea background. Naturally, the interferometric feature vector for the sea background is expressed as $\mathbf{X}_s=[\,\xi_s,\ \varphi_s\,]^{\mathrm{T}}$.

In Fig. 3, a direct measurement for the target-to-background dissimilarity can be obtained by the pth power of the feature vector distance between the target and the sea background, i.e., $\|(\mathbf{X}-\mathbf{X}_s)\|^p$. However, the interferometric magnitude and phase are completely different dimensions and may be improperly exploited in this direct dissimilarity measurement. The reasonable way to measure the target-to-background dissimilarity in the 2-D feature plane should first transform the IMP to the same dimension. Based on the investigations of the patch similarity in [46], a reciprocal form of the Gaussian kernel function is used for this purpose, and the complex interferometric dissimilarity is then defined as follows:

$$\begin{aligned}\mathrm{Dis}&=\exp(\|(\mathbf{X}-\mathbf{X}_s)\odot\mathbf{h}\|^p),\quad p=1\\&=\exp(|\xi-\xi_s|/h_{\xi_s})\exp(|\varphi-\varphi_s|/h_{\varphi_s})\end{aligned} \tag{9}$$

where $\odot$ denotes the Hadamard product, $\mathbf{h}=[\,1/h_{\xi_S},\ 1/h_{\varphi_S}\,]^{\mathrm{T}}$ denotes the bandwidth vector, the bandwidth parameters $h_{\xi S}$ and $h_{\varphi S}$ are set globally as the square roots of the normalized interferometric magnitude variance and the interferometric phase variance form the sea background, respectively, i.e., $h_{\xi_s}=(1/K\sum_{k=1}^{K}(\xi_k-\xi_s)^2)^{1/2}$ and $h_{\varphi_s}=(1/K\sum_{k=1}^{K}(\varphi_k-\varphi_s)^2)^{1/2}$, and p is set to 1 here, representing the 1 norm, i.e., $\|\mathbf{X}\|^1=\sum_i|x_i|$.

In (9), the complex interferometric dissimilarity Dis is able to properly measure the difference degree between the two dimension-different features thanks to the utilization of the bandwidth vector, which works as a normalized processing. Seen from (9), this complex interferometric dissimilarity is equal to the product of two independent components. For the magnitude one, $\exp(|\xi-\xi_s|/h_{\xi_s})$ reflects the backscattering intensity difference between the target and the sea background. The second one, i.e., $\exp(|\varphi-\varphi_s|/h_{\varphi_s})$, stands for their radial velocity difference. A longer feature distance, i.e., the interferometric magnitude distance $|\xi-\xi_s|$ or the interferometric phase distance $|\varphi-\varphi_s|$, would lead to a larger value in the corresponding dissimilarity. As to the variety properties of the sea background, the fluctuation of the interferometric feature for the background is measured by the bandwidth parameter, and this parameter is then utilized to control the dissimilarity sensitivity to the corresponding feature distance. For instance, when the distribution of sea's interferometric magnitude is more centralized with a smaller variance, the bandwidth h_{ξ_S} would be accordingly smaller. Thus, the dissimilarity term in the interferometric magnitude, i.e., $\exp(|\xi-\xi_s|/h_{\xi_s})$, would be more sensitive to the interferometric magnitude distance $|\xi-\xi_s|$ between the target and this sea background.

The definition of the above complex interferometric dissimilarity is to facilitate active marine target detection. Consider that the intensity of a metal ship is usually higher than that of sea clutter in the SAR images under the moderate sea states. Let $\xi_{\mathrm{tar}}\gg\xi_s$ in the complex interferometric dissimilarity in (9), and then the target detection metric is constructed as follows:

$$T=\exp((\xi-\xi_s)/h_{\xi_s})\exp(|\varphi-\varphi_s|/h_{\varphi_s})=\zeta\cdot\gamma\underset{H_0}{\overset{H_1}{\gtrless}}\eta \tag{10}$$

where $0<T<\infty$, $\zeta=\exp((\xi-\xi_s)/h_{\xi_s})$, and $\gamma=\exp(|\varphi-\varphi_s|/h_{\varphi_s})$ denote the IMP terms in the proposed target detection metric, respectively, H_0 denotes the null hypothesis that no moving ship exists, H_1 denotes the

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alternative hypothesis that a moving ship appears, and η denotes the detection threshold.

3) Discussions on the Proposed Metric: ATI-SAR has the advantages for low MDV in MTD missions. Based on the previous endeavors in this field, ATI detection performance, currently, is limited by several key factors, such as the reduced SCR caused by target motion (migrating pixels in SAR images), the ambiguous velocity or interferometric phase after ATI processing, and the complex detection background, i.e., the varied sea backgrounds. Refocusing the target pixels during the SAR imaging process is a useful way to improve the target SCR [33], [4]. Solutions to the last two issues are provided in the proposed method, and, in the following, the detailed discussion is given in comparison with other three current detectors, i.e., the SIMP metric [36], [37], the GP–NF, and the INF [38]. The SIMP metric is expressed as $\xi(1-\cos(\varphi-\varphi_s))$. Both the GP–NF and the INF are formulated by the same form $1/(1+\mathrm{RedR}/[\xi(1-\cos(\varphi-\varphi_s))])^{1/2}$ for ATI-SAR systems after mathematical derivations [38], where RedR denotes the adjust parameter independent of the target detection performance.

a) Blind-speed problem: Along-track separation provides phase change information, called as ATI phase, related to the radial velocities for the imaged objects. Generally, with a larger along-track space, the ATI phase is more sensitive to the radial velocity, and ATI detection can get lower MDV. The SIMP metric, the GP–NF, and the INF increase the target-to-background contrast by joining the interferometric magnitude with the interferometric phase filter, i.e., $(1-\cos(\varphi-\varphi_s))$, to suppress the sea clutter. They can improve target detection performance compared with ATI detection. However, under larger along-track separation conditions, the interferometric phase ambiguities are more sever after ATI processing, and the phenomenon leads to increased blind-speed zones for active marine target detection when using these detection methods. To deal with the bind speed problem, the proposed method uses the form of $\gamma = \exp(|\varphi-\varphi_s|/h_{\varphi s})$ for the interferometric phase filtering. The responses for the two different filters in the interferometric phase domain is shown in Fig. 4, where the channel coherence magnitude ρ of the sea clutter and the sea's average interferometric phase φ_s are set to 0.75 and 0.1 rad, respectively. For the targets exhibiting the interferometric phases approaching 0.1 rad, or corresponding to the blind speeds, the filter response in the proposed target detection metric almost becomes 1, i.e., $\varphi \to \varphi_s$, $\gamma = \exp(|\varphi-\varphi_s|/h_{\varphi s}) \approx 1$, and then this target detection metric regresses into a magnitude-based test as $T = \exp((\xi-\xi_s)/h_{\xi s})$. If the targets have sufficiently large SCR, they can be successfully detected by using the magnitude-based test. In contrast, the filter response at the blind-speed situation becomes 0 for the SIMP method and the INF because of $\varphi \to \varphi_s$, $(1-\cos(\varphi-\varphi_s)) \approx 0$. Then, the test values of these two methods, i.e., $\xi(1-\cos(\varphi-\varphi_s))$ and $1/(1+\mathrm{RedR}/[\xi(1-\cos(\varphi-\varphi_s))])^{1/2}$, would be extremely small no matter how much their interferometric magnitude is for the blind-speed targets. As a result, the two detection methods fail to identify those targets. In this sense, the proposed target detection metric makes an improvement for active marine target detection by avoiding the blind-speed problem in the single-frequency and dual-channel ATI-SAR. What is more, it can be seen from Fig. 4 that the proposed target detection metric can enhance the target-to-background contrast for the targets outside the blind-speed zone using the interferometric phase filter.

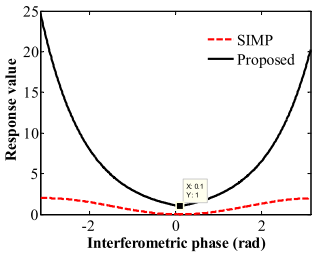


Fig. 4. Responses of the interferometric phase filters in the compared methods.

b) Adaptabilities to the varied sea backgrounds: Sea scatters have temporal velocities and differ from the ground clutter in the interferogram. The SIMP method and the INF exploit the average interferometric phase of the sea clutter to describe the main motion characteristics of the sea background and thus to determine their filter responses in the interferometric phase. In fact, this average value cannot accurately reflect the motion property of the sea clutter, because the interferometric phases are actually fluctuating around the average phase up and down over a range depending on the sea surface sate. The adaptabilities to the sea clutter background are much improved in the proposed detection metric using the IMP averages and their variances (or the bandwidth parameters) for an improvement. In particular, the average values of the IMP are utilized to determine the filter notch positions on the IMP, respectively. In addition, the notch widths of the filters depend on the corresponding bandwidth parameter. The notch widths of the IMP can change with the varied sea backgrounds and adjust the sensitivity of the proposed target detection metric to the corresponding interferometric feature adaptively.

The graphical explanation on the adaptive properties of the proposed target detection metric is provided by the following simulations. For different sea clutter regions, the probability density functions (pdfs) of the normalized n-look interferometric magnitude are shown in Fig. 5(a) (the pdf is shown in [20, eq. (11)]), where the larger the shape parameter υ, the more homogeneous the region is. For these simulation results, the channel coherence is fixed at $\rho = 0.8$. The magnitude terms, i.e., ζ, in the proposed target detection metric for these sea clutter regions are shown in Fig. 5(b). From Fig. 5(a) and (b), one can see that the magnitude term ζ is increased more quickly, as the normalized interferometric magnitude goes up for the more homogeneous sea background. This tendency is controlled by the interferometric magnitude bandwidth $h_{\xi s}$ in the proposed detection metric. For example, in the more homogeneous sea backgrounds, the bandwidth $h_{\xi s}$ is smaller in the magnitude test ζ and thus makes the magnitude test more sensitive to the interferometric magnitude.

The pdfs of the n-look interferometric phase are shown in Fig. 6(a). The diverse interferometric phase distribution

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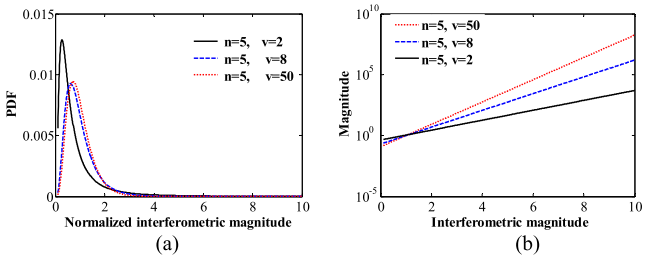


Fig. 5. (a) Pdfs of the normalized interferometric magnitude for different sea clutter regions. (b) Interferometric magnitude term of the proposed target detection metric for different sea clutter regions.

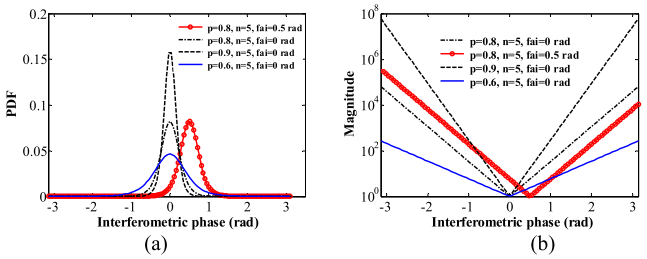


Fig. 6. (a) Pdfs of the interferometric phase for different sea backgrounds. (b) Interferometric phase term of the proposed target detection metric for different sea backgrounds.

centers of "fai" and varied dual-channel coherence magnitude stand for different sea backgrounds. The interferometric phase terms in the proposed target detection metric, i.e., γ, are accordingly created for the backgrounds and shown in Fig. 6(b). One can see that γ can attain a maximum when the interferometric phase is away from the phase center and tends to 1 when it is close to the center. In Fig. 6(a), the distribution of the interferometric phase tends to be more diffuse for the sea background with a smaller ρ. The more fluctuant interferometric phase in the background corresponds to a larger bandwidth parameter $h_{\varphi s}$ in the proposed target detection metric, and the increased bandwidth parameter enables a wider filter notch for suppressing the more spreading sea clutter, as shown in Fig. 6(b).

The proposed target detection metric is the product of the above-mentioned two terms, i.e., $T = \zeta \cdot \gamma$. Under the varied sea clutter backgrounds, the values of the target detection metric are exhibited in the 2-D interferometric feature plane in Fig. 7(a)–(c). It can be observed that the foregoing properties in each term still appear in the product and be productively realized to improve the adaptabilities of the proposed metric to the varied sea backgrounds.

B. CFAR Detection Application

1) Statistical Analysis in the Sea Background: The physical characteristics of sea surface would change with weather conditions. The constant false-alarm rate (CFAR) property is necessary for marine target detection in the varied sea clutter background for real radar systems. For this purpose, the statistics of the proposed target detection metric under the hypothesis of H_0 are discussed in this section.

In terms of physical interpretation, sea clutter in SAR images is modulated by the form of sea surface waves and

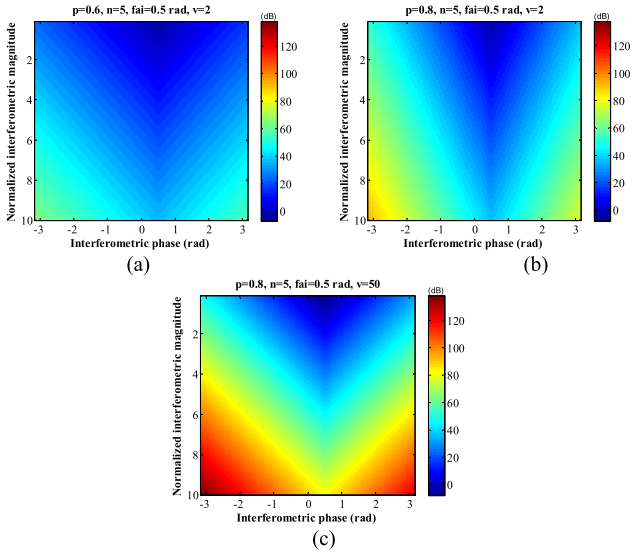


Fig. 7. Proposed target detection metric in different backgrounds displayed in decibels in the 2-D interferometric feature plane. (a) $\rho = 0.6$, $n = 5$, fai $= 0.5$ rad, and $\upsilon = 2$. (b) $\rho = 0.8$, $n = 5$, fai $= 0.5$ rad, and $\upsilon = 2$. (c) $\rho = 0.8$, $n = 5$, fai $= 0.5$ rad, and $\upsilon = 50$.

appears to be heterogeneous because of the backscattering characteristics of sea scatters and the SAR imaging mechanism [32], [47]. The statistical property of complex SAR images is usually described by the product Gauss model, that is,

$$z_i = a_i x_i, \quad a_i > 0 \tag{11}$$

where z_i denotes the complex SAR image for the ith channel, $i \in \{1, 2\}$, a_i represents the backscattering radar cross section (RCS) component, and x_i indicates the speckle component with complex standard normal distribution, i.e., $x_i \sim N_c(0, 1)$, with the mean of 0 and the variance of 1 [20], [48].

In (11), sea clutter is modeled by a statistically independent multiplicative speckle component x_i modulated by an underlying RCS component a_i. When the RCS component is described by the square root of the Gamma distribution, the intensity distribution of sea clutter will follow the K-distribution [20], [48], [49]. The K-distribution is generally used for sea clutter. However, ship wake waves caused by moving ships exist on the sea surface, and sometimes they are imaged as bright or dark pixel groups in the SAR image, greatly increasing the heterogeneous degree of the sea background. As a result, the classical K- and Weibull-distributions may fail to properly fit the sea background condition. Frery *et al.* [48] extended the K- and Weibull-distributions to fit a highly heterogeneous background by modeling the RCS component a_i with a square root of a generalized inverse Gaussian distribution. Based on this model, the variable $W := a_i^2$ is distributed with inverse chi-square distribution

$$p(W) = \frac{\upsilon^{\alpha}}{\Gamma(\alpha)} \left(\frac{1}{W}\right)^{\alpha+1} \exp\left(-\frac{\upsilon}{W}\right) \tag{12}$$

where α denotes the degrees of freedom, i.e., $\alpha > 0$, υ is a shape parameter, i.e.,$\upsilon > 0$, and $\Gamma(\cdot)$ denotes the gamma function. The degree of the heterogeneity can be measured

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with the estimated value of the shape parameter. If the value is estimated over two different areas, then the area with the larger value is more homogeneous.

Compared with the $G\Gamma D$ [50], the Log-normal [31], and the K-distribution [20], Frery's extended model in (12), also called as the P-distribution, is a general form. It is the generalized form of both the K-distribution [20] and the Gaussian distribution. More importantly, it can fit more heterogeneous backgrounds as well. Thus, Frery's extended model is adopted here to analyze the statistics of the proposed target detection metric in the varied sea backgrounds. By making the transformation $\beta = \ln(T)$, and following the derivation in Appendix A, the accurate pdf of β is given by (21), which is complex and thus intractable for a CFAR application. An approximate solution is introduced here when the IMP are assumed to be statistically independent. Under the assumption, we have the pdf of β as (25) according to Appendix A. Substituting β with $\ln(T)$ in (25), the approximate pdf of T can be obtained, that is,

$$f_T(t) = \frac{1}{t} f_\beta(\ln(t)) = \int_0^{\min[\pi/h_{\varphi s}, \beta+\xi_s/h_{\xi_s}]} \times \frac{2h_{\varphi s} h_{\xi_s} \omega^n ((\ln(t)-\delta)h_{\xi_s} + \xi_s)^{n-1}}{\mathrm{B}(n,\upsilon)(1+\omega((\ln(t)-\delta)h_{\xi_s} + \xi_s))^{n+\upsilon}} \frac{1}{t} \times \left(\begin{array}{l} \dfrac{\Gamma(n+1/2)(1-\rho^2)^n \rho\cos(h_{\varphi s}\delta)}{2\sqrt{\pi}\Gamma(n)(1-\rho^2\cos^2(h_{\varphi s}\delta))^{(n+1/2)}} \\ +\dfrac{(1-\rho^2)^n}{2\pi} {}_2F_1(n,1;1/2;\rho^2\cos^2(h_{\varphi s}\delta)) \end{array} \right) d\delta. \tag{13}$$

In (13), one can compute and record the values of the following part in the max integration domain $[0,\ \pi/h_{\varphi s}]$, that is,

$$\left(\begin{array}{l} \dfrac{\Gamma(n+1/2)(1-\rho^2)^n \rho\cos(h_{\varphi s}\delta)}{2\sqrt{\pi}\Gamma(n)(1-\rho^2\cos^2(h_{\varphi s}\delta))^{(n+1/2)}} \\ +\dfrac{(1-\rho^2)^n}{2\pi} {}_2F_1(n,1;1/2;\rho^2\cos^2(h_{\varphi s}\delta)) \end{array} \right) \tag{14}$$

and then these values can be used to efficiently compute (13) with respect to different t. Compared with the accurate one in (21), (25) can greatly reduce the computational time of the CFAR application. Then, the approximate pdf makes the CFAR application of the proposed target detection metric available. In fact, the real joint pdf in (18) suggests that, in general, the IMP is not statistically independent. Nevertheless, we provide reasonable justification for the assertion that $f_{\xi,\varphi}(\xi,\varphi) \approx f_\xi(\xi) f_\varphi(\varphi)$ by the experiments on NASA/JPL's real data (see Section IV-B).

2) CFAR Detection Scheme: Based on the definition of CFAR detection, the detection threshold η of the proposed metric under a given Pfa, i.e., P_{fa}, is determined by

$$1 - P_{\mathrm{fa}} = \int_0^\eta f_T(t)dt. \tag{15}$$

Once the background-dependent parameters, i.e., $\hat{\rho}$, $\hat{n}$, $\hat{\upsilon}$, $\hat{\alpha}$, $h_{\xi S}$, and $h_{\varphi S}$, are computed or estimated from the sea clutter background (see Appendix B), the approximate pdf of the detection metric is definitely expressed, i.e., (13), and the detection threshold η can be computed by (15) under the given Pfa with the help of the numerical solution, following with the CFAR detection.

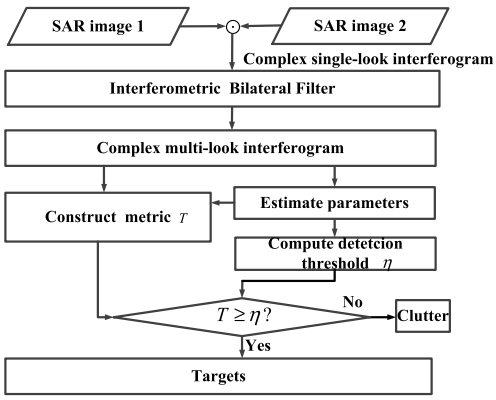


Fig. 8. Flowchart of the proposed detection approach.

The detailed CFAR detection procedures of the proposed method are summarized in the following. After the SAR imaging of the dual-channel radar returns, the essential balancing of two SAR images should be performed to counteract the unavoidable channel imperfections and imbalances [24]. With the balanced SAR images, the particular detection process is shown in Fig. 8. The complex single-look interferogram is acquired by the product of the first-channel SAR image and the conjugate of the second one. Then, the interferometric bilateral filter is applied to estimate the complex multi-look interferogram. Based on the estimation results, the approximate pdf of the proposed target detection metric T can be obtained by using (25) and (13), followed with the numerical calculation of the detection threshold under the given Pfa according to (15). During the threshold computation, the proposed target detection test for the scene can be created. Finally, the pixels whose test values exceed the threshold would be detected as the moving targets.

IV. Experimental Results and Analysis

We perform the simulations and the experiments based on the L-band real data to test the proposed method. These data are collected by NASA/JPL's airborne platform (known as AirSAR) in November 5, 1998, with the standard dual-channel ATI mode [24]. The names of the flight line and the data set are KeyWestATI-132-1 and at0043, respectively. The large scene is in the open ocean near Key West, FL, USA, and its IMP are displayed in Fig. 9(a) and (b), respectively. The spatial resolution of this image is 3.331 m $\times$ 3.898 m (range $\times$ azimuth). In Fig. 9, we use three subimages with azimuthal and range coordinate intervals [4000, 5500] and [800, 1800], [600, 1600] and [700, 1000], and [5000, 6000] and [100, 800], respectively, and these three subimages are marked by L1, L2, and L3, respectively. The main system parameters corresponding to the data are listed in Table I. According to the phase-to-velocity conversion ratio, the unambiguous radial velocity interval is computed in the range of $\pm$1.3382 m/s, because the interferometric phase is only unambiguous in $[-\pi,\ \pi]$ [19]. The interferometric phase is very sensitive to the radial

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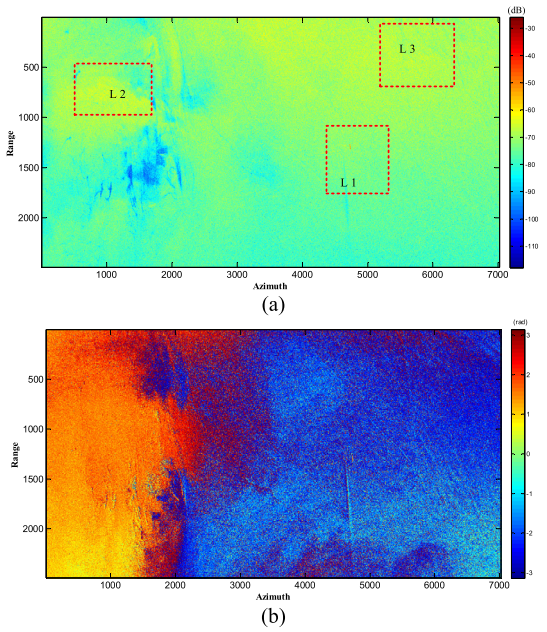


Fig. 9. Interferogram of a large scene of the open ocean near Key West, FL, USA. (a) Interferometric magnitude displayed in decibels. (b) Interferometric phase displayed in radians.

TABLE I
SYSTEM PARAMETERS OF THE SCENE ON KEYWESTATI-132-1 AT0043

Parameter	*Value*	*Units*
Radar altitude	8005.129	m
Chirp bandwidth	40×10^6	Hz
Platform velocity	216.5459	m/s
Physical baseline length	9.8	m
Radar wave length	0.242257	m
Polarization	VV	
Phase to velocity conversion ratio	0.425976	(m/s)/rad
The normal PFR	800	Hz

velocities of imaged objects. Low MDV can be available in ATI detection, but sever ATI velocity ambiguities would cause frequent blind-speed zones in company with the conventional ATI-based detection.

A. Experimental Results of Interferometric Bilateral Filter

Experimental results and discussion of the proposed preprocessing, i.e., the interferometric bilateral filtering, are provided based on the subimage L1 in this section. The IMP for L1 are shown in Fig. 10(a) and (b), respectively. In the filtering, the two parameters, i.e., h_s and h_r, are set as 10% of the window length according to the size of the minimum inscribed rectangle in a common ship image and 10^{-5} based on the variance of the background interferometric vector distance, respectively. The IMP after the filtering with a multi-look window of 5×5 pixels (range $\times$ azimuth) are shown in Fig. 10(c) and (d), respectively. It can be observed from the results that the speckle noises are dramatically reduced with the filtering.

For a better glimpse of the details on the target, the local images of the moving ship before and after the filtering are

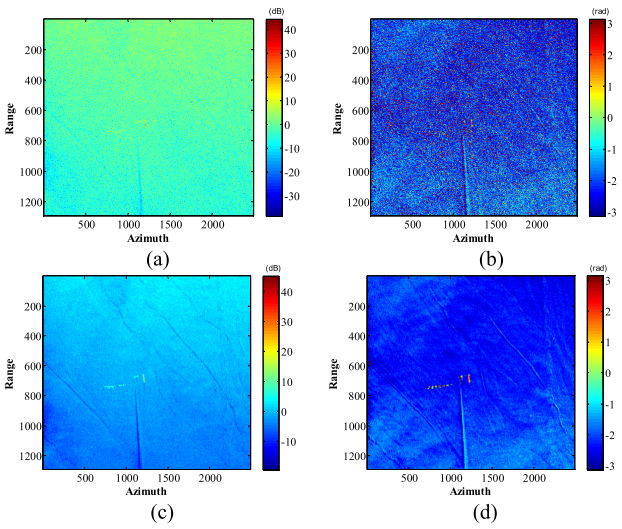


Fig. 10. Interferogram for L1. (a) Interferometric magnitude in decibels. (b) Interferometric phase in radians. (c) Interferometric magnitude in decibels and (d) interferometric phase in radians with the interferometric bilateral filter of a spatial window of 5×5 looks, respectively.

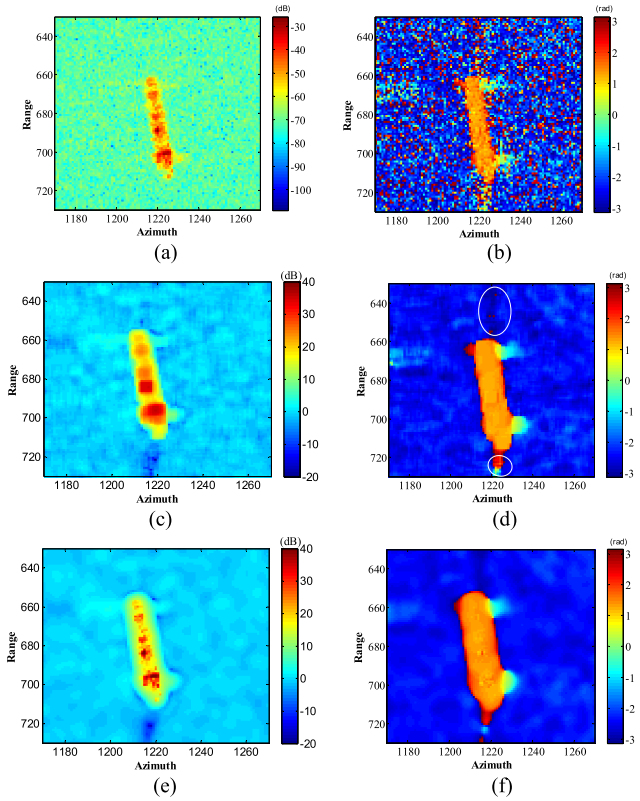


Fig. 11. Local interferometric images of the moving ship. (a) Interferometric magnitude and (b) interferometric phase before filtering. (c) Interferometric magnitude and (d) interferometric phase after the average filtering. (e) Interferometric magnitude and (f) interferometric phase after the interferometric bilateral filter.

shown in Fig. 11(a), (b), (e), and (f), accordingly. For comparisons, the IMP acquired by averaging pixels in the normal rectangle window of the same size, called as the average filter here, are displayed in Fig. 11(c) and (d), respectively. Seen from these local pictures, the noises are greatly smoothed by the filtering process. Of the two filters, the interferometric

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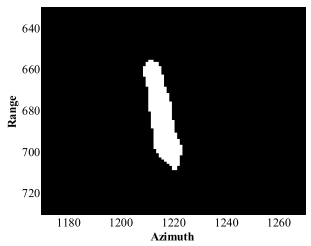


Fig. 12. Ground truth of the moving ship in the subimage L1.

bilateral filter works more properly, because the phase estimation errors with the average filter, circled by the ellipses in white in Fig. 11(d), are reduced by using the interferometric bilateral filter, as shown in Fig. 11(f). Although both these two filters cause a decline in image resolution, the interferometric bilateral filter keeps the reflectivity characteristics and the edges of the ship hull much better when smoothing the speckles.

In marine target detection, we more concern about the difference degree between the sea background and the moving ship in the complex interferometric domain before and after the filtering. The ground truth of the moving ship is acquired by the manual segmentation and shown in Fig. 12, which includes 338 target pixels. With the marks of the target pixels in the subimage L1, the IMP histograms of the sea background and the ship can be separately collected, and the corresponding histograms before and after different filtering are displayed in Fig. 13(a)–(f), respectively. Compared with the original ones, distribution differences between the sea clutter and the ship are much increased both on the IMP after applying the two filters. Of the filtered results, the interferometric phase distributions of the sea background and the ship are more centralized by using the interferometric bilateral filter. In this sense, the interferometric bilateral filter is more helpful for marine target detection.

B. CFAR Detection on Complex Mutli-Look Interferogram

After the interferometric bilateral filtering, the complex mutli-look interferogram can be available. To test the CFAR detection of the proposed method, the statistical models to describe the multi-look IMP and the real and approximate pdfs for the proposed target detection metric are verified in order, and then the CFAR detection of the proposed method is exemplified.

The complex interferogram of the sea clutter significantly presents different characteristics in different sea surface conditions. An effective statistical model of the sea background should suit for different sea surface conditions. The subimages L2 and L3 are in different regions in the large scene and with different sea surface conditions based on the visual inspection in the complex interferogram. These two subimages are used to test the fitness of the statistical models, i.e., the interferometric magnitude pdf in (22) and the interferometric phase pdf in (23). Then, the assertion that $f_{\xi,\varphi}(\xi,\varphi) \approx f_{\xi}(\xi) f_{\varphi}(\varphi)$ is tested, and proofs on the approximate pdf, i.e., (25), are given. The original IMP of the subimage L2 is shown in Fig. 14(a) and (b), respectively. After using the interferometric bilateral

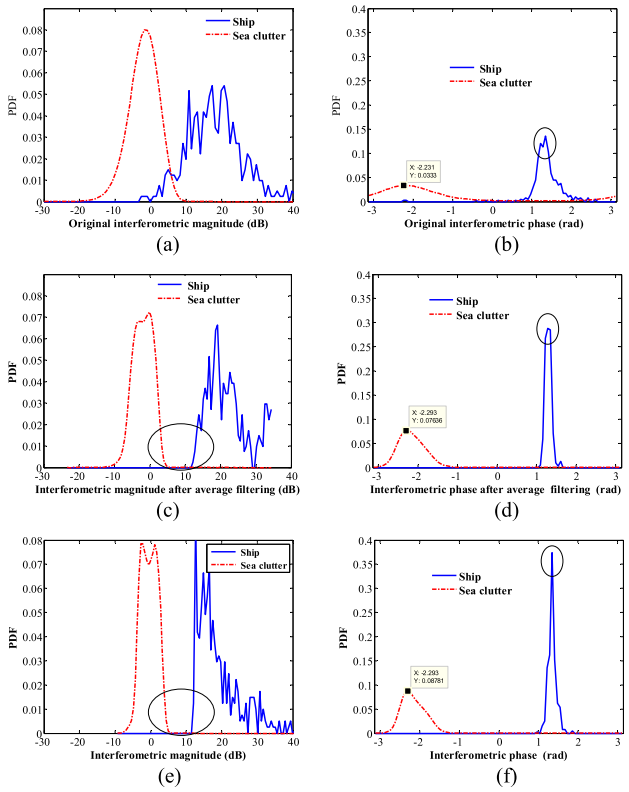


Fig. 13. The IMP histograms of the sea background and the moving ship in different methods. (a) Original interferometric magnitude and (b) original interferometric phase histograms. (c) Interferometric magnitude and (d) interferometric phase histograms after applying the average filter. (e) Interferometric magnitude and (f) interferometric phase histograms after applying the interferometric bilateral filter.

TABLE II
PARAMETER-ESTIMATED RESULTS

Parameter	L2 (5×5 -looks)	L2 (7×7 -looks)	L3 (5×5 -looks)	L3(7×7 -looks)
$\hat{\rho}$	0.9229	0.9229	0.8478	0.8478
$\hat{n}$	4.2100	5.9300	1.8200	2.5000
$\hat{v}$	6.4427	5.3458	13.9470	6.6589

filter with a spatial window of 5×5 looks along the range and azimuth directions, the multi-look IMP can be acquired, and the proposed target detection metric is then created. The statistical parameter estimates on the subimage L2 are listed in Table II. The fitting results of the IMP with the spatial window of 5×5 looks are shown in Fig. 15(a) and (b), respectively. Then, the joint histogram of the interferometric phase and magnitude and its approximate pdf, i.e., $f_{\xi,\varphi}(\xi,\varphi) \approx f_{\xi}(\xi) f_{\varphi}(\varphi)$, is shown in Fig. 15(c) and (d), respectively. In the following, the fitting results of β with the real pdf (21) and the approximate pdf (25) are shown in Fig. 11(e) and (f), respectively. Fig. 16(a)–(f) shows the corresponding results for the subimage L2 when a spatial window of 7×7 looks is adopted in the interferometric bilateral filtering. The original IMP for the subimage L3 is shown in Fig. 17(a) and (b), respectively. The same processes on the subimage L3 are executed, and the fitting results are displayed in Figs. 18 and 19.

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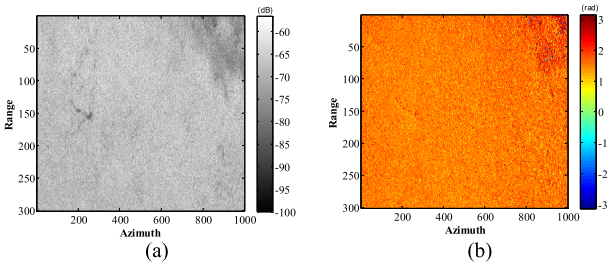


Fig. 14. Interferogram for L2. (a) Interferometric magnitude in decibels. (b) ATI in radians.

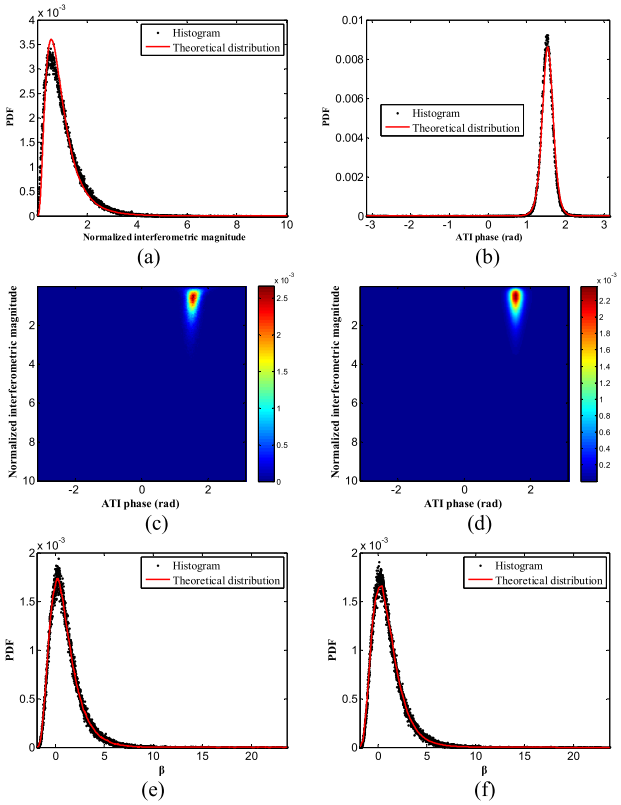


Fig. 15. Fitting results of L2 with a spatial window of 5×5 looks. (a) Normalized interferometric magnitude. (b) ATI phase. (c) Joint histogram of the IMP. (d) Approximate joint pdf of theIMP. (e) Fitting results of β with the real pdf. (e) Fitting results of β with the approximate pdf.

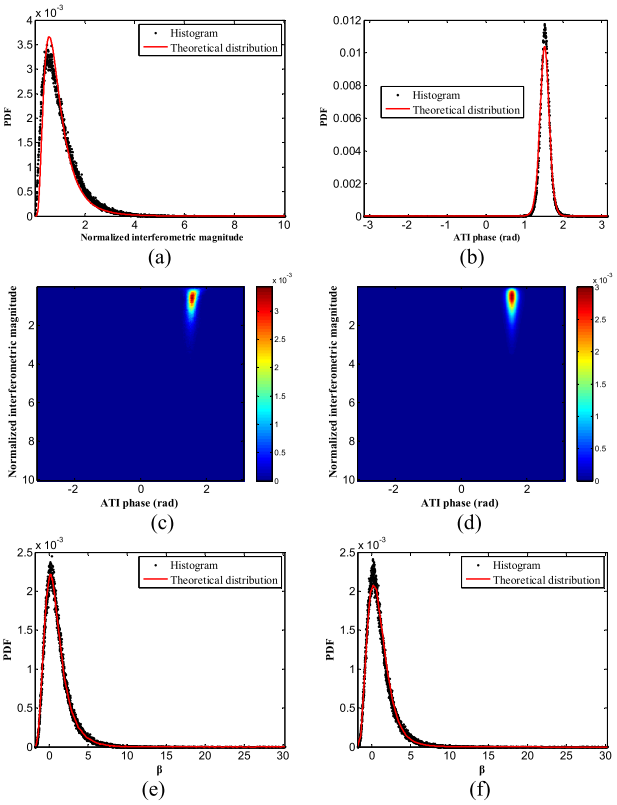


Fig. 16. Fitting results of L2 with a spatial window of 7×7 looks. (a) Normalized interferometric magnitude. (b) ATI phase. (c) Joint histogram of the IMP. (d) Approximate joint pdf of theIMP. (e) Fitting results of β with the real pdf. (e) Fitting results of β with the approximate pdf.

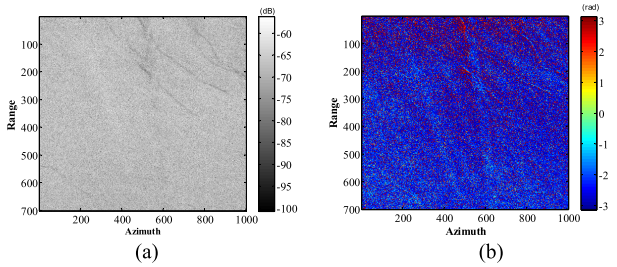


Fig. 17. Interferogram for L3. (a) Interferometric magnitude in decibels. (b) ATI in radians.

From Table II, it can be observed that the parameter $\hat{\upsilon}$ of L3 is larger compared with that of the subimage L2 under the same multi-look window. It shows that the subimage L3 is more homogeneous than the subimage L2. As the size of the multi-look window increases, the effective number of samples increases.

To evaluate the pdf modeling accuracy, we compute the mean square error (MSE) between the sample distribution $p(x_n)$ and the theoretical probability distribution $f(x_n)$, i.e., $\text{MSE} = 1/N \sum_{n}^{N} (f(x_n) - p(x_n))^2$. The smaller the MSE, the better the goodness of fit (GoF). The MSE values of the real pdf, i.e., (21), and the approximate pdf, i.e., (25), of β under different conditions are given in Table III. From the fitting results, one can observe that the used statistical models are in good agreement with the histograms of the IMP in the two different data sets. Based on the comparisons of the joint histogram of the IMP and its approximate pdf, i.e., $f_\xi(\xi) f_\varphi(\varphi)$, the assumption of $f_{\xi,\varphi}(\xi, \varphi) \approx f_\xi(\xi) f_\varphi(\varphi)$ may be close to the real relationship. Furthermore, by evaluating MSEs of the real and approximate pdfs of β, one can see that the derived approximate pdf for the transforming of the proposed metric, i.e., β, based on the assumption, fits the measured data sets as well. Thus, the assumption of $f_{\xi,\varphi}(\xi, \varphi) \approx f_\xi(\xi) f_\varphi(\varphi)$ can be acceptable for an efficient CFAR application of the proposed metric.

In the following, we perform the CFAR detection with the derived statistics on the subimage L1, because this image contains a moving ship. Detection results of the ATI detection [20], the SIMP metric [36], [37], the INF [38], [58], and the proposed metric in (10) are provided for detection comparison. To be fair, these results are all performed on the subimage

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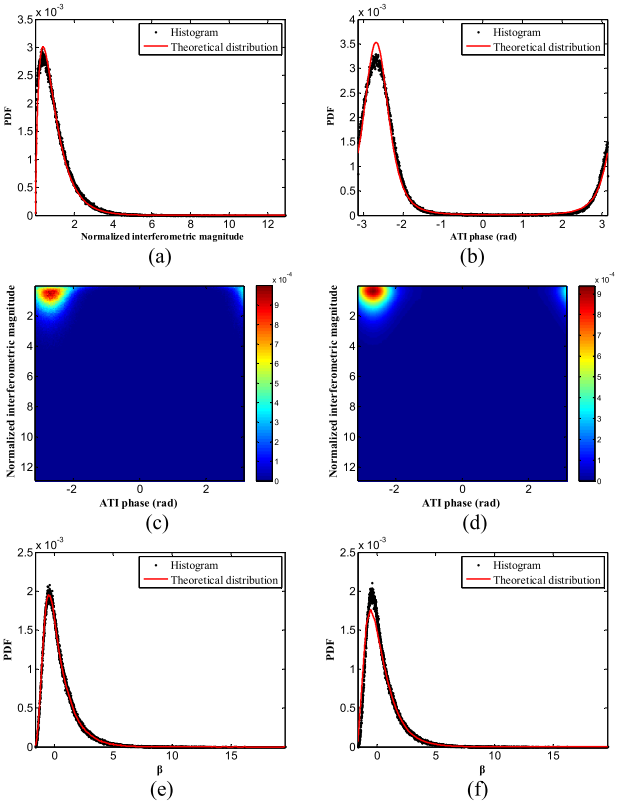


Fig. 18. Fitting results of L3 with a spatial window of 5 × 5 looks. (a) Normalized interferometric magnitude. (b) ATI phase. (c) Joint histogram of the IMP. (d) Approximate joint pdf of theIMP. (e) Fitting results of β with the real pdf. (e) Fitting results of β with the approximate pdf.

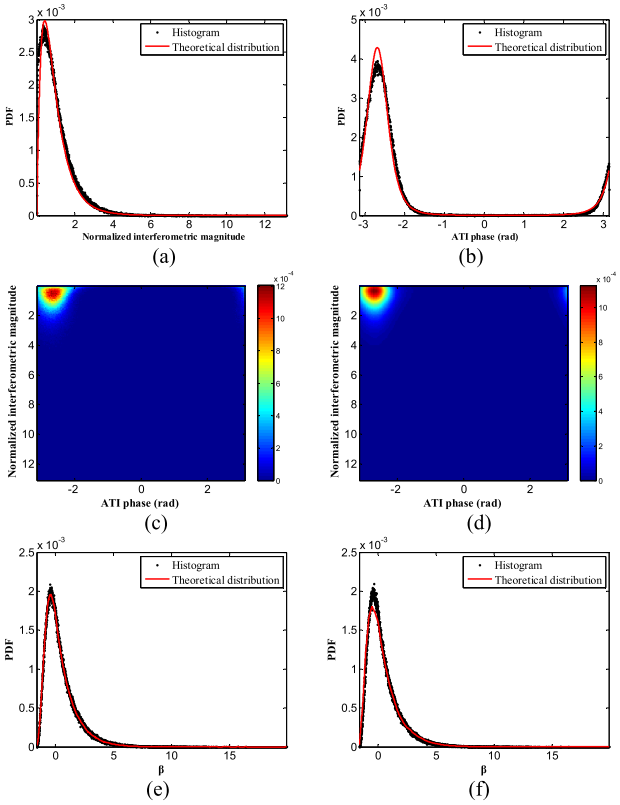


Fig. 19. Fitting results of L3 with a spatial window of 7 × 7 looks. (a) Normalized interferometric magnitude. (b) ATI phase. (c) Joint histogram of the IMP. (d) Approximate joint pdf of the IMP. (e) Fitting results of β with the real pdf. (e) Fitting results of β with the approximate pdf.

TABLE III
MSE FOR DIFFERENT IMAGES

MSE	L2 (5×5 -looks)	L2 (7×7 -looks)	L3 (5×5 -looks)	L3(7×7 -looks)
Equation (21)	$7.8368*10^{-10}$	$8.2400*10^{-10}$	$2.9198*10^{-10}$	$2.9720*10^{-10}$
Equation (25)	$1.3220*10^{-9}$	$2.2372*10^{-9}$	$3.7957*10^{-9}$	$1.7452*10^{-9}$

L1 with the interferometric bilateral filter. In this subscene, the sea's average interferometric phase φ_s is estimated as -2.1458 rad, its average interferometric magnitude is normalized to 1, and the square roots of the normalized IMP variances are $\sigma_{\xi_s} = 0.4981$ and $\sigma_{\varphi_s} = 0.2616$ rad, respectively. The ATI test, the SIMP metric, the INF, and the proposed metric are created and shown in Figs. 10(d) and 20(a)–(c), respectively, where the parameter RedR is set to 1 for the INF according to [38]. It can be seen that the moving ship seems to be most detectable, and the sea background looks more suppressed in the proposed test.

Then, the fitting results of the distribution for the ATI test, the INF method, and the proposed metric (with the approximate pdf of β) for the sea clutter are displayed in Fig. 21(a)–(c) with the parameters $\hat{\rho} = 0.62$, $\hat{n} = 7.72$, and $\hat{\upsilon} = 2.01$. The SIMP method is not compared here, because its pdf is only available for single-look complex interferogram in the open literature. According to (15), the detection threshold is computed as 92.8299 ($\beta = 9.2997$) when using $P_{fa} = 10^{-6}$. The CFAR detection results are shown in Fig. 21(d)–(f). It can be seen that the moving ship in the subimage L1 can be effectively detected by using the proposed method in the CFAR approach.

We test the computational time of the compared detectors in CFAR detection based on the same complex multi-look interferogram with the processor of my computer: Intel Core i5-3570 CPU at 3.4 GHz, and the RAM is 4 GB. The time for the ATI, the INF, the proposed method with the approximate pdf, and the proposed method with the real pdf are 2.7527, 0.6310, 3.1497, and 232.4512 s, respectively. The proposed metric and the ATI method cost much more time than the INF approach, because computing the Gauss hyper geometric function for every variable in the pdfs is time consuming. It can be observed that, if we use the approximate pdf to replace the real one for the CFAR detection, our computational time is reduced by a factor of 73.8007 on the test data.

C. Detection Performance Evaluation Based on the Simulations in Real-Data Background

To quantitatively evaluate the detection performance, the receiver operating characteristic (ROC) curve of a plot of detection ratio versus different actual Pfas is adopted. The detection ratio and the actual Pfa are defined in the following. The number of sea clutter pixels in the subimage L1 is computed by $N_c = MN - N_t$, where $M$and N denote the

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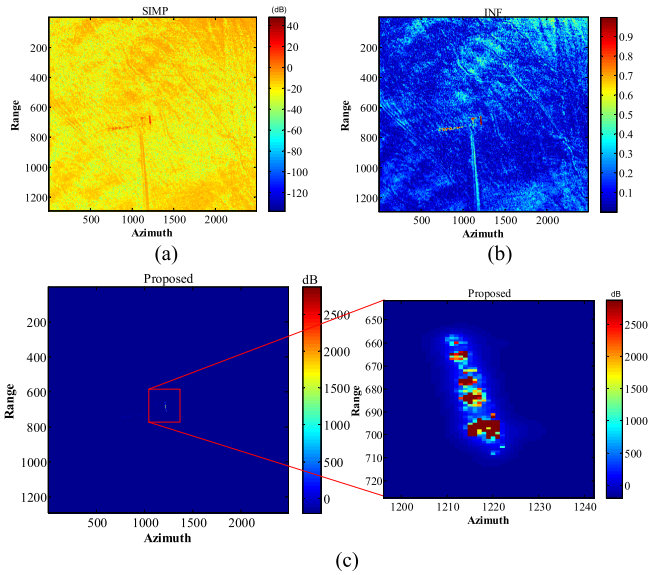


Fig. 20. Tests for (a) SIMP metric in decibels, (b) INF, and (c) proposed metric displayed in a log scale.

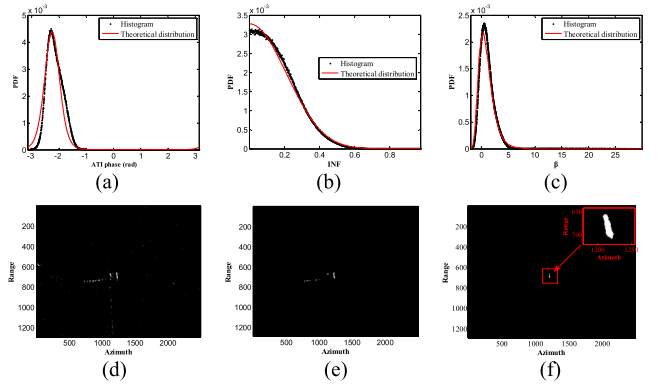


Fig. 21. Fitting results on L1 for (a) ATI test, (b) INF method, and (c) proposed metric. CFAR detection results of (d) ATI test, (e) INF method, and (f) proposed method when Pfa $= 10^{-6}$.

pixel numbers in the range and azimuthal directions in the interferogram, respectively, and N_t is the ship pixel number according to the ground truth shown in Fig. 12. Under a detection threshold, the number of detected ship pixels is denoted by $N_{\rm dt}$, and the number of detected false alarms is denoted by $N_{\rm dc}$. Then, the detection ratio is defined as follows:

$$P_d = N_{\rm dt}/N_t \tag{16}$$

and the actual Pfa is denoted by

$$P_{\rm fa} = N_{\rm dc}/N_c. \tag{17}$$

Under different actual Pfas, the detection thresholds of the compared methods can be computed according to the background histograms, since we can distinguish the clutter pixels in L1 based on the ground truth shown in Fig. 12. The detection ratios of these methods against different Pfas are shown in Fig. 22. Seen from the results, the proposed metric has a comparable performance with the SIMP metric and the INF, and a superior performance over ATI detection on the test data.

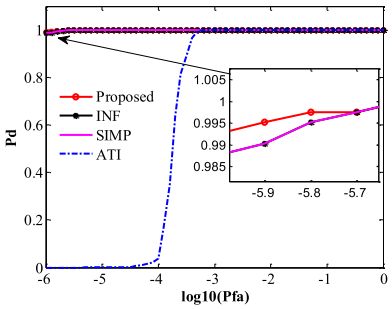


Fig. 22. ROC curves against different Pfas on the subimage L1.

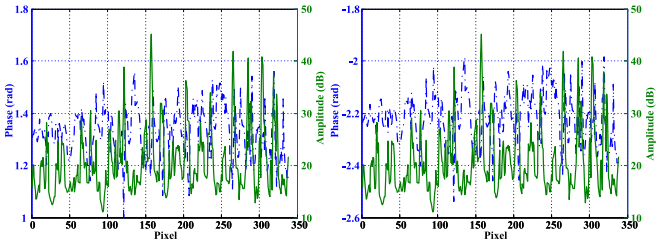


Fig. 23. (a) Normalized interferometric magnitude (solid line) and the interferometric phase (dashed line) for the real ship displayed in decibels and radians, respectively. (b) Normalized interferometric magnitude (dashed line) and phase (solid line) for the simulated ship with the interferometric phase center of −2.2 rad and an average SCR of 27.7730 dB on the interferometric magnitude.

It is significant to comprehensively evaluate the detection performance under different SCR values and varied ship radial velocities. However, in the subimage L1, there is only one ship with the specific radial velocity or interferometric phase, and SCR value. Therefore, we do simulations with the real background by changing the ship's radial velocity and the SCR value for the comprehensive analysis. According to the ground truth shown in Fig. 12, the normalized IMP for the ship pixels is shown in Fig. 23(a) with solid and dashed lines, respectively. These pixels belong to one ship, but their normalized IMP vary in the certain ranges such as the normalized interferometric magnitude in the range of 15–45 dB with an expectation of 27.7730 dB and the interferometric phases fluctuating around the center value of 1.3255 rad up and down. To reserve the target properties in the simulations, we choose to change the expectation of the normalized interferometric magnitude and the center value of the interferometric phase for the ship to represent the different SCR and radial velocity conditions, respectively. For instance, a simulated ship with a phase center of −2.2 rad, corresponding to the radial velocity of $v_r = -0.9371$ m/s, is shown in Fig. 23(b). In this way, the ROC curves against different target radial velocities under an average SCR value of 27.7730 dB are shown in Fig. 17, where Fig. 24(a) and (b) shows the results under Pfa $= 10^{-4}$ and 10^{-6}, respectively. For the sea background, we can know that the interferometric phase center is −2.1458 rad, corresponding to the radial velocity $v_r = -0.91$ m/s in the unambiguous radial velocity interval. The interferometric phase unambiguous interval of 2π leads the repeated blind speeds at $-0.91 \pm 2 \times 1.3382 \times k$ m/s, $k = 0, 1, 2, \ldots$, i.e., −0.9100 m/s, 1.7664 m/s,. . ., for the ATI detection shown

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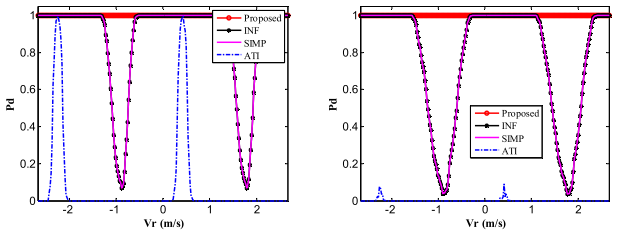


Fig. 24. ROC curves against different radial velocities when the average SCR of 27.7730 dB. (a) Results under Pfa $= 10^{-4}$. (b) Results under Pfa $= 10^{-6}$.

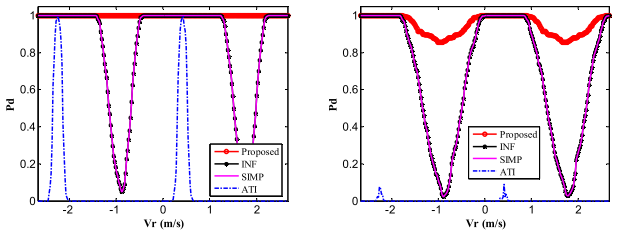


Fig. 25. ROC curves against different radial velocities when the average SCR of 25 dB. (a) Results under Pfa $= 10^{-4}$. (b) Results under Pfa $= 10^{-6}$.

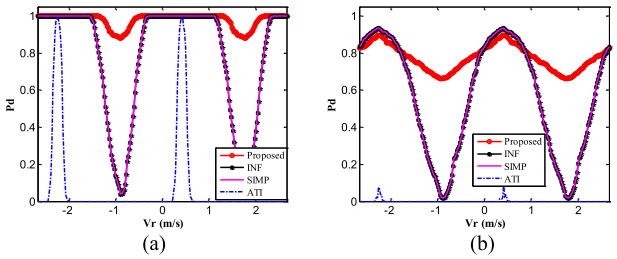


Fig. 26. ROC curves against different radial velocities when the average SCR of 23 dB. (a) Results under Pfa $= 10^{-4}$. (b) Results under Pfa $= 10^{-6}$.

in Fig. 17. The SIMP method and the INF suffer from sever detectability reduction in the blind-speed zones, but the proposed metric acquires satisfactory detection results at the situation. As the Pfa increases, the detection performance of the compared methods reduces. The Pd of the proposed metric appears to decrease near the blind-speed zones, because the proposed metric almost becomes a magnitude-based test with less contribution of the interferometric phase at these blind speeds.

Under different target SCR conditions in the simulation, the ROC curves of the compared methods versus different velocities are studied, and the results are shown in Figs. 25 and 26. The results show that the proposed method exhibits nearly comparable detection results for the targets with velocities far from the blind speeds compared with the SIMP and INF methods, and outperforms them in the blind-speed zones.

V. Conclusion

This article introduces a method for active marine target detection based on complex interferometric dissimilarity in dual-channel ATI-SAR systems. During the interferometric information measurements, an interferometric bilateral filter is designed to smooth the speckles and locally adapt to the spatial structure of the interferometric image. Then, the target detection metric is constructed according to the complex interferometric dissimilarity between the moving ships and the sea background. Different from the most existing ATI-SAR detectors, the proposed target detection metric can adaptively regress into a magnitude-based test for the targets with blind speeds and thus mitigate the blind-speed problem. Furthermore, the pdf of the proposed metric is derived based on the compound Gaussian model to realize the CFAR detection. Since the pdf is complex and of high computational complexity, we use an approximate pdf to facilitate the CFAR detection. The experiments on the data collected by NASA/JPL's AirSAR and several simulations are performed to test the proposed method and the GoF for the approximate pdf. It is conclusive that the interferometric bilateral filter is effective in smoothing noise and restoring interferogram edge, and it can further increase the interferogram histogram differences between targets and sea clutters to facilitate the following detection. The approximate pdf is in good agreement with the histograms of the real data and greatly reduced the computational time compared with the accurate pdf. The CFAR detection of the proposed method is therefore verified. In marine target detection, the proposed target detection metric exhibits obvious superiority over the compared methods for the targets moving with the blind speeds.

In addition, a certain issue about the usefulness of the proposed target detection metric still remains open. The proposed metric can ease the target blind-speed detection problem. However, the metric would be sensitive to the noisy speckles with large magnitude due to its phase filtering property. With the current complex multi-look interferometric techniques, the large-magnitude speckles and some interferogram's measurement errors can be largely reduced, and thus the proposed metric properly performs on the test data. In our future work, we would like to study its detection ability with more data sets from different SAR platforms.

Appendix A

In [20], the joint pdf of the IMP for the heterogeneous background is derived as follows:

$$\begin{aligned} f_{\xi,\varphi}(\xi,\varphi) &= \frac{(2n)^{2n}\upsilon^{\alpha}\Gamma(2n+\alpha)\Gamma(\alpha+2)}{2\sqrt{\pi}\,\Gamma(n)(1-\rho^2)^{-n-\alpha}\Gamma(\alpha)\Gamma(n+\alpha+3/2)} \\ &\quad \cdot \xi^{2n-1}((1-\rho^2)\upsilon - 2n\xi(\rho\cos(\varphi-\varphi_s)-1))^{-2n-\alpha} \\ &\quad \times {}_2F_1\left(\begin{matrix} 2n+\alpha, n-1; n+\alpha+\dfrac{3}{2}; \\ \dfrac{(1-\rho^2)\upsilon - 2n\xi(\rho\cos(\varphi-\varphi_s)+1)}{(1-\rho^2)\upsilon - 2n\xi(\rho\cos(\varphi-\varphi_s)-1)} \end{matrix}\right) \end{aligned} \quad (18)$$

where n denotes the number of independently distributed samples taken to estimate the multi-look complex interferogram (also called as the effective number of looks), ρ denotes the channel coherence magnitude, and ${}_2F_1(\cdot)$ denotes the Gauss hyper geometric function.

By making the transformations of $\delta = |\varphi - \varphi_s|/h_{\varphi s}$ and $\kappa = (\xi - \xi_s)/h_{\xi s}$, and using the substitution of

$$g_0 = \frac{(2n)^{2n}\upsilon^{\alpha}\Gamma(2n+\alpha)\Gamma(\alpha+2)}{\sqrt{\pi}\,\Gamma(n)(1-\rho^2)^{-n-\alpha}\Gamma(\alpha)\Gamma(n+\alpha+3/2)}$$

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(18) is

$$
\begin{aligned}
&f_{\kappa,\delta}(\kappa,\delta)\\
&= g_0 h_{\varphi_s} h_{\xi_s}((1-\rho^2)\upsilon - 2n(\kappa h_{\xi_s}+\xi_s)\\
&\quad\times(\rho\cos(h_{\varphi_s}\delta)-1))^{-2n-\alpha}\\
&\quad\times {}_2F_1\left(\begin{array}{l}2n+\alpha, n-1; n+\alpha+\dfrac{3}{2};\\ \dfrac{(1-\rho^2)\upsilon - 2n(\kappa h_{\xi_s}+\xi_s)(\rho\cos(h_{\varphi_s}\delta)+1)}{(1-\rho^2)\upsilon - 2n(\kappa h_{\xi_s}+\xi_s)(\rho\cos(h_{\varphi_s}\delta)-1)}\end{array}\right)
\end{aligned}
\tag{19}
$$

With the transformation of $\kappa = \beta - \delta$, (19) is By integrating (20), as shown at the bottom of this page, with respect to the variable δ, the pdf of β can be acquired by where $\kappa \geq -\xi_s/h_{\xi_s}$, and thus $0 \leq \delta \leq \min[\pi/h_{\varphi_s}, \beta + \xi_s/h_{\xi_s}]$.

Seen from (21), as shown at the bottom of this page, it is complex. An approximate solution is acquired by using the assumption of $f_{\xi,\varphi}(\xi,\varphi) \approx f_\xi(\xi) f_\varphi(\varphi)$. Integrating (18) with respect to the interferometric phase leads to an approximate form for the interferometric magnitude density function, that is,

$$
f_\xi(\xi) \cong \frac{\omega^{\hat{n}}}{\mathrm{B}(\hat{n},\upsilon)} \frac{\xi^{\hat{n}-1}}{(1+\omega\xi)^{\hat{n}+\upsilon}} \tag{22}
$$

where $\omega = 2n/(\alpha(1+\rho))$ and $\mathrm{B}(\cdot)$ is the Beta function. The density function in (22) is called as the polynomial or P-distribution [20], which has been widely used for describing the statistical property of the interferometric magnitude in heterogeneous backgrounds.

The pdf of the interferometric phase under the assumption of Gaussian backscatter is shown to agree almost perfectly for the extended model [20], which is

$$
\begin{aligned}
f_\varphi(\varphi) = &\frac{\Gamma(n+1/2)(1-\rho^2)^n \rho\cos(\varphi-\varphi_s)}{2\sqrt{\pi}\Gamma(n)(1-\rho^2\cos^2(\varphi-\varphi_s))^{(n+1/2)}}\\
&+\frac{(1-\rho^2)^n}{2\pi} {}_2F_1(n,1;1/2;\rho^2\cos^2(\varphi-\theta))
\end{aligned}
\tag{23}
$$

Based on $f_{\xi,\varphi}(\xi,\varphi) \approx f_\xi(\xi) f_\varphi(\varphi)$, (19) can approximate to

$$
\begin{aligned}
&f_{\kappa,\delta}(\kappa,\delta)\\
&= \frac{2h_{\varphi_s} h_{\xi_s} \omega^n (\kappa h_{\xi_s}+\xi_s)^{n-1}}{\mathrm{B}(n,\upsilon)(1+\omega(\kappa h_{\xi_s}+\xi_s))^{n+\upsilon}}\\
&\quad\times\left(\begin{array}{l}\dfrac{\Gamma(n+1/2)(1-\rho^2)^n\rho\cos(h_{\varphi_s}\delta)}{2\sqrt{\pi}\Gamma(n)(1-\rho^2\cos^2(h_{\varphi_s}\delta))^{(n+1/2)}}\\ +\dfrac{(1-\rho^2)^n}{2\pi} {}_2F_1(n,1;1/2;\rho^2\cos^2(h_{\varphi_s}\delta))\end{array}\right)
\end{aligned}
\tag{24}
$$

and the pdf of β is therefore as follows:

$$
\begin{aligned}
f_\beta(\beta) = &\int_0^{\min[\pi/h_{\varphi_s}, \beta+\xi_s/h_{\xi_s}]}\\
&\times\frac{2h_{\varphi_s} h_{\xi_s} \omega^n ((\beta-\delta)h_{\xi_s}+\xi_s)^{n-1}}{\mathrm{B}(n,\upsilon)(1+\omega((\beta-\delta)h_{\xi_s}+\xi_s))^{n+\upsilon}}\\
&\times\left(\begin{array}{l}\dfrac{\Gamma(n+1/2)(1-\rho^2)^n\rho\cos(h_{\varphi_s}\delta)}{2\sqrt{\pi}\Gamma(n)(1-\rho^2\cos^2(h_{\varphi_s}\delta))^{(n+1/2)}}\\ +\dfrac{(1-\rho^2)^n}{2\pi} {}_2F_1(n,1;1/2;\rho^2\cos^2(h_{\varphi_s}\delta))\end{array}\right) d\delta.
\end{aligned}
\tag{25}
$$

APPENDIX B

In Frery's extended model, the parametric estimators were already derived by [20], [48], and [51]–[57]. The relevant contents are introduced here for applying the CFAR detection of the proposed method. The estimators for the dual-channel coherence magnitude ρ, the effective number of looks $\hat{n}$, the shape parameter υ, and the parameter α are given in the following, i.e., $\hat{\rho}$, $\hat{n}$, $\hat{\upsilon}$, and $\hat{\alpha}$, respectively.

According to [55], the ML estimate of the dual-channel coherence magnitude is expressed as follows:

$$
\hat{\rho} = \frac{2\sum_{k=1}^{K}(|I_{1,k}|+|I_{2,k}|)}{\sum_{k=1}^{K}\sqrt{|I_{1,k}|\cdot|I_{2,k}|}\cos(\theta_{1,k}-\theta_{2,k}+\hat{\varphi}_s)} \tag{26}
$$

where $I_{1,k}$ and $\theta_{1,k}$ denote the intensity and phase of the fore channel for the pixel k, respectively, and $I_{2,k}$ and $\theta_{2,k}$ are those

$$
\begin{aligned}
f_{\beta,\delta}(\beta,\delta) = &\, g_0 h_{\varphi_s} h_{\xi_s}((\beta-\delta)h_{\xi_s}+\xi_s)^{2n-1}\\
&\times((1-\rho^2)\upsilon - 2n((\beta-\delta)h_{\xi_s}+\xi_s)(\rho\cos(h_{\varphi_s}\delta)-1))^{-2n-\alpha}\\
&\times {}_2F_1\left(\begin{array}{l}2n+\alpha, n-1; n+\alpha+\dfrac{3}{2};\\ \dfrac{(1-\rho^2)\upsilon - 2n((\beta-\delta)h_{\xi_s}+\xi_s)(\rho\cos(h_{\varphi_s}\delta)+1)}{(1-\rho^2)\upsilon - 2n((\beta-\delta)h_{\xi_s}+\xi_s)(\rho\cos(h_{\varphi_s}\delta)-1)}\end{array}\right).
\end{aligned}
\tag{20}
$$

$$
\begin{aligned}
f_\beta(\beta) = &\, g_0 h_{\varphi_s} h_{\xi_s}\int_0^{\min[\pi/h_{\varphi_s}, \beta+\xi_s/h_{\xi_s}]}((\beta-\delta)h_{\xi_s}+\xi_s)^{2n-1}\\
&\times((1-\rho^2)\upsilon - 2n((\beta-\delta)h_{\xi_s}+\xi_s)(\rho\cos(h_{\varphi_s}\delta)-1))^{-2n-\alpha}\\
&\times {}_2F_1\left(\begin{array}{l}2n+\alpha, n-1; n+\alpha+\dfrac{3}{2};\\ \dfrac{(1-\rho^2)\upsilon - 2n((\beta-\delta)h_{\xi_s}+\xi_s)(\rho\cos(h_{\varphi_s}\delta)+1)}{(1-\rho^2)\upsilon - 2n((\beta-\delta)h_{\xi_s}+\xi_s)(\rho\cos(h_{\varphi_s}\delta)-1)}\end{array}\right) d\delta
\end{aligned}
\tag{21}
$$

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of the aft channel. Another most common estimator is

$$\hat{\rho} = \left| \sum_{k=1}^{n} z_1(k) z_2(k)^* \right| \Big/ \sqrt{\sum_{k=1}^{n} z_1(k) z_1(k)^* \sum_{k=1}^{n} z_2(k) z_2(k)^*}$$

in [20].

A fast and robust estimator for the effective number of looks, i.e., $\hat{n}$, based on the method of higher moments possessing [53], [57] is given here. Let Φ be the real part of the exponential of the interferometric phase φ, i.e., $\Phi = Re(\exp(j\varphi))$, which can be efficiently estimated by taking the real part of the normalized complex pixels in the multi-look interferogram. In [57], the estimation of the second moment of Φ is acquired by averaging K multi-look pixels, i.e., $\hat{I}_2$, and formulated by

$$\hat{I}_2 - \frac{(1-\hat{\rho}^2)^{\hat{n}} + (\hat{n}\hat{\rho}^2 - 1)}{(\hat{n}-1)\hat{\rho}^2} = 0 \tag{27}$$

The estimate of $\hat{n}$ can be acquired by finding the root of (27).With the method of moments (MoM), an estimate to the shape parameter, i.e., $\hat{\upsilon}$, is

$$\hat{\upsilon} = \frac{2\hat{n}m_2 - (\hat{n}+1)m_1^2}{\hat{n}m_2 - (\hat{n}+1)m_1^2} \tag{28}$$

where m_r denotes the rth moment of ξ and is calculated by

$$m_r := E(\xi^r) = \int_0^{+\infty} \xi^r f_\xi(\xi)\, d\xi. \tag{29}$$

For the parameter α, it can be estimated by

$$\hat{\alpha} = \frac{2\,\hat{n}\,m_1\,\mathrm{B}(\hat{n},\hat{\upsilon})}{(1+\hat{\rho})\,\mathrm{B}(\hat{n}+1,\hat{\upsilon}-1)}. \tag{30}$$

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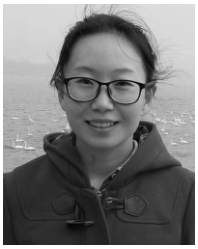
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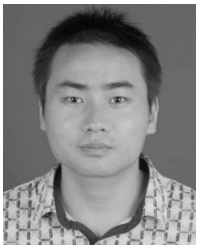
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